We know that $\Phi_B = BA\cos\theta$

$= 40{,}773.9 \times 10^{-9} \times 3 \times \cos 43^{o}$

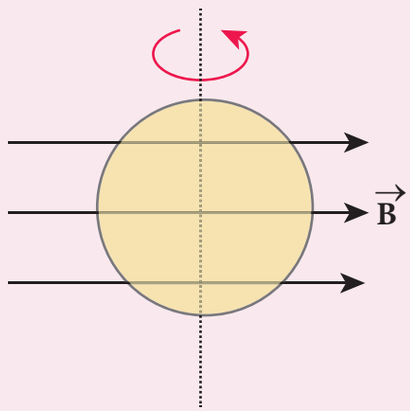
$= 40.7739 \times 10^{-6} \times 3 \times 0.7314$

$= 89.47 \times 10^{-6}$ Wb

$\Phi_B = 89.47\,\mu\text{Wb}$

EXAMPLE 4.2

A circular loop of area $5 \times 10^{-2}\text{m}^2$ rotates in a uniform magnetic field of 0.2 T. If the loop rotates about its diameter which is perpendicular to the magnetic field as shown in figure. Find the magnetic flux linked with the loop when its plane is (i) normal to the field (ii) inclined 60° to the field and (iii) parallel to the field.



Solution

$A = 5 \times 10^{-2}\,\text{m}^2$; $B = 0.2$ T

(i) $\theta = 0^{\circ}$;

$\Phi_B = BA\cos\theta = 0.2\times5\times10^{-2}\times\cos0^{\circ}$

$\Phi_B = 1\times10^{-2}Wb$

(ii) $\theta = 90^{\circ} - 60^{\circ} = 30^{\circ}$;

$\Phi_B = BA\cos\theta = 0.2\times5\times10^{-2}\times\cos30^{\circ}$

$\Phi_B = 1\times10^{-2}\times\frac{\sqrt{3}}{2} = 8.66\times10^{-3}Wb$

(iii) $\theta = 90^{\circ}$;

$\Phi_B = BA\cos90^{\circ} = 0$

4.1.3 Faraday's Experiments on Electromagnetic Induction

First Experiment

Consider a closed circuit consisting of a coil *C* of insulated wire and a galvanometer *G* as shown in Figure 4.2(a). The galvanometer does not indicate deflection as there is no electric current in the circuit.

When a bar magnet is inserted into the stationary coil, with its north pole facing the coil, there is a momentary deflection in the galvanometer. This indicates that an electric current is set up in the coil (Figure 4.2(b)). If the magnet is kept stationary inside the coil, the galvanometer does not indicate deflection (Figure 4.2(c)).

The bar magnet is now withdrawn from the coil, the galvanometer again gives a momentary deflection but in the opposite direction. So the electric current flows in opposite direction (Figure 4.2(d)). Now if the magnet is moved faster, it gives a larger deflection due to a greater current in the circuit (Figure 4.2(e))

The bar magnet is reversed i.e., the south pole now faces the coil. When the above experiment is repeated, the deflections are opposite to that obtained in the case of north pole (Figure 4.2(f)).

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